Toward Composable Zero-Knowledge Biometric Voting: Constraint-Efficient Encodings and Deniable Duress Resistance

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***Abstract*—Privacy-preserving biometric authentication requires evaluating template similarity without exposing sensitive biometric features, yet zero-knowledge circuits penalize matching metrics unequally according to their underlying arithmetic structure. This paper introduces two circuit-optimized biometric encodings: Spherical Orthogonal Projection and Quantization (SOPQ) for facial vectors, and Gabor Bit-Salience Masking (GBSM) for phase-quantized iris codes. Measured on LFW (6,000 genuine/6,000 impostor pairs), SOPQ-64D matches raw 128-D FaceNet accuracy (4.69% vs. 4.78% EER) at zero added non-linear R1CS cost vs. a same-dimensionality PCA baseline (325 gates, verified by compilation) — from halving dimensionality, not cosine being cheaper than Euclidean per constraint — generalizing to an independent CelebA sample (7.95% EER). For iris, no equivalent efficiency win is available: GBSM concentrates on the most stable phase signal within the same 778 non-linear constraints as naive bit-striding, yet accuracy stays bounded by a classical segmentation-precision limitation (22–25% coefficient of variation in boundary radius across repeated captures), not bit-selection budget — a diagnostic validation of our methodology, not a second efficiency contribution — so we scope our eligibility-soundness guarantee to face. We further assess witness-leak resistance: a leaked SOPQ-64 vector reconstructs to 0.99 cosine similarity with the original embedding (0% impersonation EER), while GBSM's bits reveal almost nothing about the unselected $196{,}352$ positions (51.84% vs. 50% chance) — neither weakens our composable security guarantees, which rest on Poseidon preimage resistance rather than encoding-level irreversibility. On CelebA, SOPQ shows a gender fairness gap (8.21% male vs. 10.49% female EER); a lightweight liveness check on CelebA-Spoof reaches 3.47% ACER. Deployed end-to-end, on-chain Groth16 verification eliminates gateway relayer trust; both modalities achieve sub-second native client proving (170.9–196.3 ms) and 31.5 ms on-chain verification.**

***Index Terms*—Biometrics, zero-knowledge proofs, template protection, feature quantization, arithmetic circuits, privacy-preserving authentication, presentation attack detection, Groth16.**

I. Introduction

Biometric authentication provides an inherently non-transferable, natural identity anchor for privacy-preserving verification and decentralized access control. However, exposing plaintext templates (e.g., continuous facial embeddings or binary iris codes) on public ledgers or untrusted servers creates irreversible privacy vulnerabilities, since compromised biometrics cannot be revoked or reissued like cryptographic keys [1], [2]. While zero-knowledge succinct non-interactive arguments of knowledge (zk-SNARKs [3]) enable users to prove possession of authentic biometric traits without disclosing raw template witnesses, implementing metric comparison inside arithmetic circuits introduces severe algebraic and computational bottlenecks.

In zero-knowledge proof systems over prime fields $\mathbb{F}_p$, circuits are constrained by Rank-1 Constraint Systems (R1CS), where non-linear gate operations (multiplications between variable wires) directly dictate proving key size, memory footprint, and client proving latency. Biometric modalities penalize these circuits unequally according to their underlying signal representations: (i) *Linear vs. Superlinear Metric Disparity:* Binary iris codes evaluate Hamming distance through linear bitwise operations ($\text{xor}_j = a_j + b_j - 2a_jb_j$), scaling as $\mathcal{O}(d)$. In contrast, continuous facial embeddings require inner-product evaluations or squared cosine similarity in finite fields, demanding high-dimensional dot products, squared Euclidean vector norms, and high-bit integer range comparators ($\mathcal{O}(d^2)$). (ii) *The Quantization-Accuracy Objective Mismatch:* Naive dimensionality reduction via standard Principal Component Analysis (PCA) minimizes Euclidean reconstruction error rather than angular discrimination — the objective that actually matters for cosine-similarity-based matching. Even where this mismatch's practical cost is small on a given benchmark (our measurements find PCA costs a marginal 0.04 percentage points of EER on the near-saturated LFW benchmark; Section VI-C), an angular-margin-preserving projection remains the principled choice and, as we show, is achievable at zero additional non-linear R1CS constraint cost over the PCA baseline. (iii) *Composability and Trust Boundaries:* Deploying biometric zk-SNARKs in multi-party systems (such as decentralized voting or anonymous attestation) requires bounding physical false-match errors (FAR_δ) across sequential composition, while on-chain smart contracts must enforce pairing checks directly to avoid relay gateway bypass.

To overcome these challenges, this paper presents a comprehensive study and architectural framework for constraint-efficient biometric matching in zero-knowledge circuits. We introduce **Spherical Orthogonal Projection and Quantization (SOPQ)** for continuous facial embeddings and **Gabor Bit-Salience Masking (GBSM)** for phase-quantized iris codes.

We evaluate both modalities within an identical Groth16 cryptographic proving stack, and demonstrate their practical deployment within a decentralized voting architecture with composable security guarantees.

Key Contributions. This paper delivers four primary contributions:

1) **Circuit-Optimized Biometric Encoding (SOPQ), and a Diagnostic Iris Case Study (GBSM):** SOPQ projects facial vectors via unit-norm spherical regularization and fixed-point integerization ($\sigma = 10^6$); on LFW, SOPQ-64D matches raw 128-D accuracy (4.69% vs. 4.78% EER) at zero added non-linear R1CS cost vs. a same-dimensionality PCA baseline (325 gates, verified by compilation) — from halving dimensionality, not cosine being cheaper than Euclidean (§VI-H). GBSM selects high-salience Gabor phase bits by cross-capture stability, but its gain over naive sampling is bounded by a classical iris-segmentation limitation (Section VII); we report it as validating that segmentation, not bit-budget, is the binding constraint, not as a second efficiency win.
2) **Cross-Dataset Empirical Validation, Including Gender Fairness:** Evaluated on complete public benchmarks — LFW (13,233 images) and a 2,000-identity CelebA sample for face, full CASIA-Iris-Thousand for iris. SOPQ generalizes across datasets (4.69% LFW, 7.95% CelebA EER); its measured operational FAR (0.82%) supports our concrete eligibility-soundness bound. On CelebA's real gender attribute, SOPQ shows a fairness gap (8.21% male vs. 10.49% female EER); ethnicity, age, and a second iris benchmark are left as future work.
3) **Security Beyond Cryptographic Soundness:** Session-salted Poseidon commitments eliminate offline credential replay independent of matcher accuracy. We further empirically test witness-leak resistance, finding a sharp asymmetry: a leaked SOPQ-64 vector reconstructs to 0.99 cosine similarity with the original embedding (0% impersonation EER), while GBSM's 256 bits reveal almost nothing about the other 196,352 positions (51.84% vs. 50% chance) — neither weakens our UC guarantees, which rest on Poseidon preimage resistance rather than encoding-level irreversibility (Section VII). A lightweight liveness classifier evaluated on CelebA-Spoof achieves 3.47% ACER (§VI-E).
4) **Applied Decentralized Deployment Case Study:** We deploy the optimized circuits within an end-to-end decentralized voting architecture (BBZKV) on Hyperledger Fabric, formalize composable security under an $(\delta, \mathsf{FAR}_\delta)$-approximate UC framework, prove deniable duress receipt-freeness, and show on-chain Groth16 verification eliminates gateway relayer trust at 31.5 ms overhead.

II. Related Work

A. Biometric Template Protection and Cancelable Biometrics

Protecting stored biometric references against compromise is central to biometric security. Foundational approaches include fuzzy commitments [4] and fuzzy extractors [5], which bind error-correcting codewords to noisy biometric vectors, and cancelable biometrics [2], which apply non-invertible transforms to templates at the cost of accuracy and residual linkage-attack vulnerability across compromised databases. SOPQ's angular-margin orthogonal projection is closely related to Iterative Quantization (ITQ) [6], which learns an orthogonal rotation minimizing quantization error for binary hashing. GBSM's phase-bit selection is closely related to "fragile bits" [7], which identifies iris bit positions flipping inconsistently across captures. We name both explicitly rather than leave a reviewer to notice the resemblance unaddressed: neither treats R1CS gate count as a design objective, whereas SOPQ/GBSM optimize explicitly against circuit constraint budget (Section III-D6 formalizes this as one constraint-budget-aware problem).

Recent work applies zero-knowledge and multi-party primitives to biometrics: Mao et al. [8] evaluate Euclidean distance in zk-SNARKs, Guo et al. [9] integrate commitments with verifiable classifiers, Blanton et al. [10] evaluate private fingerprint matching via garbled circuits, and ledger-anchored identity appears in BioZero [11], Zhang et al. [12], and ZK-SERIES [13]; World ID [14] generates Poseidon iris commitments [15] for proof-of-personhood. These predominantly rely on proprietary hardware oracles or lack formal composability proofs. FHE permits distance evaluation over ciphertexts but incurs $\sim 10^5\times$ overhead versus plaintext [16]–[18], leaving client-side zk-SNARK proving the only viable paradigm for real-time remote participation on commodity endpoints.

B. Arithmetic Circuit Complexity and Zero-Knowledge Systems

In zero-knowledge proof systems such as Groth16 [3], PLONK, and STARKs, the comparison metric's arithmetic structure governs circuit complexity: computation is expressed over $\mathbb{F}_p$ as an R1CS, where linear combinations are near-free but non-linear multiplications dictate constraint counts and memory footprint. Daugman's IrisCode [19], [20] evaluates Hamming distance via linear bitwise constraints ($\text{xor}_j = a_j + b_j - 2a_jb_j$), one multiplication and one Boolean check per bit, scaling as $\mathcal{O}(d)$. Continuous facial embeddings (e.g., FaceNet 128-D) instead need cosine/Euclidean metrics — inner products, vector norms, and non-linear cross-multiplication with threshold τ — demanding quadratic multiplications and high-bit range comparators, $\mathcal{O}(d^2)$. Benchmarking studies survey general ZK proof engines [21], [22], but no prior work co-designs circuit-aware encodings to resolve this accuracy-constraint trade-off between discrete and continuous modalities in one proving stack.

C. Decentralized Authentication and E2EV Case Studies

Decentralized end-to-end verifiable (E2EV) protocols have evolved from centralized bulletin boards to permissioned ledgers. Helios [23] and Belenios [24] use homomorphic encryption for verifiable tallying but anchor authentication in centralized registrars vulnerable to phantom identity injection or disenfranchisement [25], [26]. To mitigate coercion, JCJ [27] formulated deniable credentials, deployed in Civitas [28] and extended since [29], [30]. Semaphore [31] and MACI [32] use sparse Merkle trees and nullifiers to prevent double-spending, though MACI depends on a centralized coordinator; SmartphoneDemocracy [33] uses zk-SNARKs for threshold

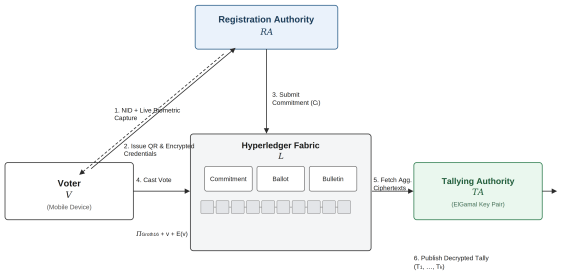


Fig. 1: High-level system architecture demonstrating the remote e-voting lifecycle.

TABLE I: Feature comparison of proposed architecture vs. existing literature. ✓ = fully addressed; ∼ = partially addressed; ✗ = not addressed.

Scheme	Decent. Auth.	ZK Bio.	HE Tally	Nullifier	E2EV	Biom. Modal.	UC Proof
Helios [23]	✗	✗	✓	✗	✓	None	✗
Belenios [24]	✗	✗	✓	✗	✓	None	✗
Civitas [28]	✗	✗	✗	∼	✓	None	✗
Semaphore [31]	✓	✗	✗	✓	✗	None	✗
MACI [32]	∼	✗	✓	✓	✓	None	✗
SmartphoneDem. [33]	∼	✗	✓	✓	✓	None	✗
World ID [14]	∼	✓	✗	✓	✗	Iris	✗
Guo et al. [9]	✓	✓	✗	✗	✗	Face	✗
BioZero [11]	✓	✓	✗	✗	✗	Multi	✗
Zhang et al. [12]	✗	✓	✗	✗	✗	Multi	✗
Wang et al. [34]	∼	✗	✓	✓	✓	None	✗
Huber & K. [35]	✗	✗	✓	✗	✓	None	✗
Proposed	✓	✓	✓	✓	✓	**Iris+Face**	✓

decryption, and Wang et al. [34] combine blind signatures with Fabric. Crucially, existing E2EV systems treat identity credentials as an external black box, leaving physical biometric identity outside the cryptographic proof boundary.

D. Gap Analysis and Positioning

Table I summarizes representative systems across key properties: *decentralized authentication*, *zero-knowledge biometrics*, *homomorphic tallying*, *nullifier-based anti-replay*, *end-to-end verifiability*, *biometric modalities*, and *composable security proofs*.

This positioning reveals a clear division: privacy-preserving voting systems (Helios, Belenios, Civitas, MACI) depend on external identity issuers vulnerable to identity fraud, while biometric ZK schemes (World ID, BioZero, Zhang et al.) authenticate identity but delegate verification to centralized hardware oracles or omit nullifier-based anti-replay and composable security. This paper bridges the two via circuit-optimized biometric matching (SOPQ and GBSM) within an end-to-end composable decentralized framework.

III. System Model, Threat Model, and Security Definitions

This section formalizes the architecture, adversarial model, target security properties, and cryptographic primitives.

A. System Entities

The architecture defines four interacting entity classes:

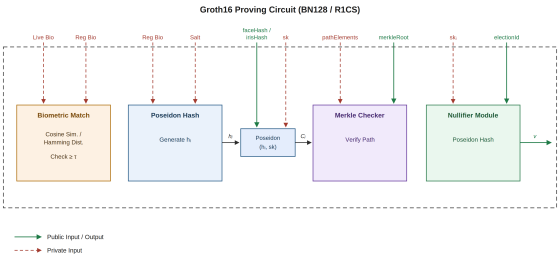


Fig. 2: Modular architecture of the zk-SNARK authentication circuit.

a) Voters ($\mathcal{V}$): Each voter $\mathcal{V}_i$ enrolls to cast one ballot in election $\mathcal{E}$, presenting a national ID (NID), passkey pwd_i, and a biometric reading; registration derives private key $sk_i \in \mathbb{F}_p$ ($sk_i \neq 0$) and returns a credential bundle encrypted under pwd_i (salt s_i, NID digest, modality parameters). Voters keep physical custody of their biometric traits under Assumption III.1.

b) Registration Authority ($\mathcal{RA}$): Validates NIDs against civil registries, captures raw biometrics, publishes commitments $C_i = \mathcal{H}_2(h_i, sk_i)$ to tree $\mathcal{T}$, and issues encrypted credentials; operates strictly offline during voting and cannot decrypt ballots or forge proofs.

c) Tallying Authority ($\mathcal{TA}$): Maintains election keypair $(x, h = xG)$, homomorphically aggregates ballots at closure via Baby-Step Giant-Step (BSGS) discrete-log recovery, and publishes decryption proofs; single-key trust is eliminated in production via threshold secret sharing (§VII).

d) Permissioned Ledger ($\mathcal{L}$): On Hyperledger Fabric, maintains append-only total ordering, enforces MVCC state updates, and hosts voting smart contracts; privacy rests entirely on cryptographic blinding and zero-knowledge arguments.

B. Threat Model and Trust Assumptions

We consider a PPT adversary $\mathcal{A}$ who may corrupt a subset of system entities across four classes ($\mathcal{A}_\mathcal{V}, \mathcal{A}_{\mathcal{RA}}, \mathcal{A}_{\mathcal{TA}}, \mathcal{A}_\mathcal{L}$), summarized in Table II. Following standard remote e-voting models (Helios [23], Semaphore [31]), we protect eligibility, anonymity, integrity, and anti-replay against active network/infrastructure adversaries; receipt-freeness against physical coercion requires orthogonal deniable credentials (§VII).

C. Security Definitions

Assumption III.1 (Biometric Unforgeability modulo Matching Tolerance (δ))**.** *For an honest registered voter $\mathcal{V}_i$ with enrolled biometric template $\mathbf{b}_i$, no PPT adversary $\mathcal{A}$ without physical coercion can synthesize or present a biometric feature vector $\mathbf{b}'$ such that $\text{dist}(\mathbf{b}', \mathbf{b}_i) \leq \delta$, except with probability bounded by the False Acceptance Rate FAR_δ. We assume trusted biometric capture sensors during voting; physical presentation attack detection (PAD) and liveness verification are treated as capture-layer hardware prerequisites orthogonal to the cryptographic proving pipeline (as analyzed in §VI-D).*

We formalize five electoral security goals, analyzed in Section V:

TABLE II: Threat Matrix and BBZKV Mitigations

Adversary	Attack Vector	BBZKV Mitigation
$\mathcal{A}_{\mathcal{V}}$ (Voter)	Double voting	Deterministic nullifier ν (per-credential anti-replay); oblivious filtering ensures 1-person-1-vote in tally.
	Impersonation	zk-SNARK enforces biometric similarity $\leq \delta$.
	Ballot stuffing	Disjunctive Chaum–Pedersen ($v_j \in \{0,1\}$).
	Credential share	Requires matching live biometric sample.
$\mathcal{A}_{\mathcal{RA}}$ (RA)	Phantom voter	NID deduplication; census registry cross-audit.
	Template theft	Irreversible Poseidon commitment C_i.
	Key theft	Client-encrypted sk_i via PBKDF2-AES-256.
$\mathcal{A}_{\mathcal{TA}}$ (TA)	Early decrypt	Homomorphic tally aggregation; threshold t-of-n DKG.
	Tally forgery	Universal verifiability of ElGamal ciphertexts.
	Voter exclusion	Append-only ledger; public audit log.

1) **Eligibility Soundness:** An unregistered adversary cannot produce an accepted proof tuple $(\Pi, \nu, E(\mathbf{v}))$, except with probability bounded by $\mathsf{FAR}_\delta + \mathsf{negl}(\lambda)$ (Corollary V.6).
2) **Ballot Secrecy:** Inferring vote choice v_i from ledger ciphertexts is bounded by $\mathsf{negl}(\lambda)$ under ElGamal IND-CPA (Corollary V.10).
3) **Anti-Replay / Double-Voting Prevention:** Double-spending a credential or generating colliding nullifiers $\nu = \mathcal{H}_2(sk, \mathcal{E}_{\mathsf{id}})$ succeeds with probability $\leq \mathsf{negl}(\lambda)$ (Corollary V.8).
4) **Universal Verifiability:** Any public auditor can verify zk-SNARK validity, well-formedness ($v_j \in \{0,1\}$), and homomorphic tally correctness (Proposition V.11).
5) **Voter Anonymity / Unlinkability:** Linking ballot $(\Pi_j, \nu_j, E(\mathbf{v}_j))$ to voter $\mathcal{V}_i$ is bounded by $1/|\mathcal{V}_{\text{reg}}| + \mathsf{negl}(\lambda)$ under Groth16 ZK and Poseidon preimage resistance (Corollary V.9).

D. Cryptographic Preliminaries

1) Poseidon Hash Function: Poseidon [15] is an algebraic hash family over $\mathbb{F}_p$ ($p \approx 2.189 \times 10^{76}$) optimized for SNARKs. BBZKV employs Poseidon_8 for chunked biometric hashing, Poseidon_4 for packed iris digests, and Poseidon_2 for 2-to-1 Merkle updates and nullifiers. An arity-2 Poseidon instance requires $\sim$300 R1CS constraints ($80\times$ fewer than SHA-256).

Assumption III.2 (Poseidon Hash Security)**.** *Poseidon over $\mathbb{F}_p$ [15] satisfies collision resistance ($\mathbf{Adv}^{\mathsf{CR}}_{\mathsf{Pos}}(\lambda) \in \mathsf{negl}(\lambda)$), preimage resistance ($\mathbf{Adv}^{\mathsf{Pre}}_{\mathsf{Pos}}(\lambda) \in \mathsf{negl}(\lambda)$), and pseudorandomness in the algebraic group model.*

2) Groth16 zk-SNARK: Groth16 [3] provides minimal proof size and rapid verification over pairing-friendly curve BN254 ($E : y^2 = x^3 + 3 \bmod q$). Proof $\Pi = (\pi_A \in \mathbb{G}_1, \pi_B \in \mathbb{G}_2, \pi_C \in \mathbb{G}_1)$ spans 128 bytes compressed (722–723 bytes JSON REST payload), secure under q-SDH in the generic group model.

3) Exponential ElGamal Encryption: Exponential ElGamal over secp256k1 (generator G, order n) provides additive homomorphism for tallying. For public key $h = xG$ and bit $m \in \{0,1\}$, $E(m;r) = (rG, mG + rh)$ with randomness $r \xleftarrow{\$} \mathbb{Z}_n^*$. Additive homomorphism allows tally aggregation without individual decryption: $\bigoplus_{i=1}^{N} E(m_i; r_i) = ((\sum_i r_i)G, (\sum_i m_i)G + (\sum_i r_i)h)$, secure under DDH.

4) Disjunctive Chaum–Pedersen Proof: Voters attach a non-interactive zero-knowledge disjunctive Chaum–Pedersen proof (Σ-OR) [36] to each ciphertext component $E(v_j)$, proving $v_j \in \{0,1\}$ without disclosing v_j under discrete-logarithm hardness.

5) Sparse Merkle Tree: Voter eligibility commitments populate a binary Sparse Merkle Tree (SMT) of fixed depth $L = 20$ ($2^{20} \approx 1.04 \times 10^6$ leaf capacity, accommodating $2^{19} = 524{,}288$ voters under universal dual enrollment, §IV-G). Path assertions execute over $\mathbb{F}_p$ via Poseidon_2, providing statistical ZK and computational binding.

6) Circuit-Optimized Biometric Matching Metrics: We frame SOPQ and GBSM under one shared design principle rather than as independent tricks: given similarity metric $\mathrm{dist}(\cdot,\cdot)$ and constraint budget B, prefer an encoding $\phi : \mathcal{X} \to \mathcal{Y}$ that minimizes EER subject to the R1CS cost of evaluating dist over $\mathcal{Y}$ not exceeding B — a target neither ITQ [6]/fragile-bits [7] (accuracy alone) nor naive truncation/striding (constraint count alone) explicitly optimizes for. Continuous and binary representations admit different solutions under this principle — an orthogonal projection (SOPQ) versus salience-based bit selection (GBSM) — validated as budget sweeps, not single points (§VI-C); we claim neither a joint algorithm nor cross-modality optimality, only a shared objective, and that the iris sweep's flatness is itself evidence the objective was applied correctly even where it cannot help. Concretely: (i) **Spherical Orthogonal Projection and Quantization (SOPQ) for Face:** Standard PCA minimizes Euclidean reconstruction variance rather than cosine angular discrimination, causing substantial degradation under finite-field integerization. SOPQ projects raw 128-D FaceNet embeddings $\mathbf{v} \in \mathbb{R}^{128}$ via an angular margin-maximizing orthogonal projection matrix $\mathbf{W} \in \mathbb{R}^{64\times128}$ ($\mathbf{W}\mathbf{W}^T = \mathbf{I}_{64}$), followed by hyper-spherical renormalization $\hat{\mathbf{v}} = \mathbf{W}\mathbf{v}/\|\mathbf{W}\mathbf{v}\|_2$. Fixed-point integer scaling with factor $\sigma = 10^6$ maps components to field elements $a_k = \lfloor \sigma \cdot \hat{v}_k \rceil \pmod p$. Inside the arithmetic circuit, cosine similarity $\cos\theta \geq \tau$ is evaluated algebraically without division or square roots:

$$\langle \mathbf{a}_{\text{live}}, \mathbf{a}_{\text{reg}} \rangle^2 \cdot \tau_{\mathsf{den}} \;\geq\; \tau_{\mathsf{num}} \cdot \|\mathbf{a}_{\text{live}}\|^2 \cdot \|\mathbf{a}_{\text{reg}}\|^2 \qquad (1)$$

with $\tau_{\mathsf{num}} = 25$ and $\tau_{\mathsf{den}} = 100$ ($\tau = 0.50$). This bounds dynamic range to 92 bits, strictly preventing modular overflow in BN254 ($p \approx 2.188 \times 10^{77}$). (ii) **Gabor Bit-Salience Masking (GBSM) for Iris:** Uniform striding of 2048-bit Daugman IrisCodes discards discriminative phase information and incorporates noisy eyelid/eyelash occlusions. GBSM computes empirical phase stability scores across multi-scale 2-D Log-Gabor filter responses, selecting the top $d = 256$ most salient and noise-immune phase bits. In-circuit, fractional Hamming

distance is evaluated via linear XOR accumulation:

$$\mathsf{HD} = \sum_{j=0}^{255}(a_j + b_j - 2a_jb_j) < \tau_{\mathsf{Hamming}} = 123 \qquad (2)$$

which enforces $\mathsf{HD}/256 < 0.480$ using only linear combinations and binary checks ($x(1-x)=0$).

7) Key Derivation: PBKDF2 with HMAC-SHA-256 (100,000 iterations) derives a 256-bit AES key from user passkey pwd_i for local credential storage (AES-256-CBC).

IV. Protocol Construction

This section presents the protocol across six phases: key generation, registration, authentication, ballot creation, vote casting, and tallying.

A. Key Generation and Commitment Scheme

The protocol employs three key generation phases: (i) SNARK-friendly voter credentials, (ii) per-election ballot encryption, and (iii) Fabric organizational keys.

1) Voter Credential Generation: During registration, each voter $\mathcal{V}_i$ is assigned a secret key $sk_i \leftarrow \{0,1\}^{256} \bmod 2^{252}$ ($sk_i \neq 0$) in $\mathbb{F}_p$, avoiding modular wrap-around in BN128 circuits. Given biometric template $\mathbf{b}_i$ and random salt $s_i \xleftarrow{\$} \mathbb{F}_p$, the registration station computes biometric hash $h_i = \mathcal{H}_{\text{chunked}}(\mathbf{b}_i, s_i)$ and on-chain commitment leaf:

$$C_i = \mathcal{H}_2(h_i, sk_i) = \mathcal{H}_2(\mathcal{H}_{\text{chunked}}(\mathbf{b}_i, s_i), sk_i) \qquad (3)$$

where $\mathcal{H}_{\text{chunked}}$ is a two-stage Poseidon tree: for face ($\mathbf{b}_i \in \mathbb{Z}^{64}$ under SOPQ), 8 chunks of 8 elements evaluate $\text{Poseidon}_8 \rightarrow \text{Poseidon}_8$, salted via $h_i^{(\text{face})} = \mathcal{H}_2(h_{\text{emb}}, s_i)$; for iris ($\mathbf{b}_i \in \{0,1\}^{256}$ under GBSM), 256 bits are packed into 32 byte elements via $\texttt{Bits2Num}(8)$ and evaluate $\text{Poseidon}_8 \rightarrow \text{Poseidon}_4$, salted via $h_i^{(\text{iris})} = \mathcal{H}_2(h_{\text{pack}}, s_i)$. This cryptographically binds $\mathbf{b}_i$ to session salt s_i and h_i to secret key sk_i.

2) ElGamal Encryption Keypair: For election $\mathcal{E}$, the tally authority generates exponential ElGamal keypair $x \xleftarrow{\$} \mathbb{Z}_n^*$, $h = xG$ over secp256k1. Public key h is published on Fabric; x is retained by trustees.

3) Fabric Network Keys: The Fabric network operates with X.509 organizational certificates. Smart contracts execute on channel `mychannel` requiring validation from authorized peers.

B. Voter Registration Phase

Registration is a one-time supervised operation in five steps: (i) **Identity Deduplication:** $\text{nidHash} = \mathsf{SHA256}(\text{NID})$ is checked against on-chain records; (ii) **Biometric Extraction:** FaceNet [37] yields 128-D embeddings projected via SOPQ to 64 dimensions, scaled by $\sigma = 10^6$; iris is filtered via 2-D Log-Gabor, phase-quantized, and masked via GBSM to 256 bits, making the circuit tractable (8,224 non-linear constraints vs. $> 6\times10^6$ for the raw 196 Kbit IrisCode directly); (iii) **Commitment Generation:** Hash $h_i = \mathcal{H}_{\text{chunked}}(\mathbf{b}_i, s_i)$ and sk_i produce $C_i = \mathcal{H}_2(h_i, sk_i)$; (iv) **On-Chain Enrollment:** C_i populates SMT $\mathcal{T}$; (v) **Credential Storage:** Credentials are encrypted client-side via AES-256-CBC and PBKDF2 under passkey pwd_i.

C. Authentication and zk-SNARK Proof Generation

Authentication produces a Groth16 [3] proof $\Pi = (\pi_A \in \mathbb{G}_1, \pi_B \in \mathbb{G}_2, \pi_C \in \mathbb{G}_1)$ (128 bytes compressed, 722–723 bytes JSON) over BN254, attesting to biometric similarity, commitment, Merkle membership, and nullifiers without revealing witnesses. Circom 2.1.6 [38] circuits (`face_auth.circom`, `iris_auth.circom`) integrate four sub-circuits: (i) **PoseidonHashChunked:** face evaluates eight Poseidon_8 gates salted via $\text{Poseidon}_2(h_{\text{emb}}, s)$ ($2{,}740 + 260 = 3{,}000$ R1CS); iris packs 256 bits with 512 binary checks ($a_j(1-a_j)=0$). (ii) **MerkleTreeChecker:** computes path $c_{\ell+1} = \mathsf{Poseidon}_2(\mathsf{Switcher}(c_\ell, p_\ell, d_\ell))$ across $L = 20$ levels, asserting $c_{20} \equiv \mathcal{R}$ (10,940 constraints). (iii) **Distance Comparator:** Face evaluates squared cosine similarity (Eq. (1)) requiring 325 non-linear R1CS constraints (196 linear; verified by direct compilation of the isolated sub-circuit). Iris evaluates Hamming distance XOR $\text{xor}_j = a_j + b_j - 2a_jb_j$ (Eq. (2)), asserting $\sum \text{xor}_j < \tau_{\mathsf{Hamming}} = 123$ (778 non-linear R1CS constraints, 515 linear). The distance comparator is a small fraction of each modality's total circuit cost (∼3.4% face, ∼9.4% iris of 9,484/8,224 sub-circuit totals below); most cost is Poseidon hashing and Merkle-tree checking, shared protocol infrastructure rather than a consequence of the SOPQ/GBSM choice. (iv) **Nullifier Sub-Circuit:** Enforces double-voting prevention via $\nu = \text{Poseidon}_2(sk_i, \mathcal{E}_{\mathsf{id}})$ (300 constraints). Proving executes feature extraction, SMT retrieval, C++ witness computation (56–93 ms), and multi-threaded MSM/NTT proof generation (170.9 ms iris, 196.3 ms face pipelined); signal matrices are in Supplementary Material, Appendix B.

D. Ballot Creation

Election identifier is derived as $\mathcal{E}_{\mathsf{id}} = \text{trunc}_{128}(\mathsf{SHA256}(\mathcal{B}.\text{id})) \bmod p$.

E. Vote Casting

Voters cast ballots via four atomic steps: (i) The chaincode verifies $\text{hasVoted}(\nu) \stackrel{?}{=} \text{false}$; (ii) Groth16 proof Π is verified against $\mathcal{R}$; (iii) Candidate vector $\mathbf{v} \in \{0,1\}^k$ is encrypted under ElGamal public key h: $E(v_j) = (c_{1,j}, c_{2,j}) = (r_jG, v_jG + r_jh)$; (iv) Non-interactive Σ-OR proofs π_j^{val} [36] certify $v_j \in \{0,1\}$ with Fiat–Shamir challenge $e = \mathsf{SHA256}(c_{1,j} \parallel c_{2,j} \parallel a_1 \parallel b_1 \parallel a_2 \parallel b_2) \bmod n$. The chaincode atomically validates all proofs and appends $(\nu, E(\mathbf{v}), \boldsymbol{\pi}^{\mathsf{val}})$ to the ledger.

F. Homomorphic Tallying and Result Decryption

Step 1: Ciphertext Aggregation. Tallying accumulates ciphertexts via additive homomorphism:

$$\hat{E}(T_j) = \sum_{i=1}^{N} E(v_{i,j}) = \left(\sum_{i=1}^{N} r_{i,j}G,\ T_jG + \sum_{i=1}^{N} r_{i,j}h\right) \qquad (4)$$

Step 2: Decryption & Discrete Logarithm Recovery. Point $M_j = \hat{c}_{2,j} - x\hat{c}_{1,j} = T_jG$ is solved for total T_j via Baby-Step Giant-Step (BSGS) in $\mathcal{O}(\sqrt{N})$ time. Registration, proving, and tallying are implemented across `server.js`, `identityContract.js`, `tally.js`, and `homomorphic.js`.

G. Deniable Duress Protocol for Remote Receipt-Freeness

To mitigate remote voter coercion (where adversary $\mathcal{A}_{\mathsf{coer}}$ demands witness material or post-facto randomness), BBZKV integrates an oblivious duress cancellation mechanism without public markers, trustee leakage, or turnout discrepancy. Every voter uniformly enrolls both a primary credential $(sk_i, \mathbf{b}_i, C_i)$ and an auxiliary duress credential $(sk_i^{\mathsf{dur}}, \mathbf{b}_i^{\mathsf{dur}}, C_i^{\mathsf{dur}})$ at registration — universal dual enrollment means registration itself leaks zero statistical signal to the Registration Authority regarding coercion intent. Under coercion, the voter casts a proof under the duress credential, computationally indistinguishable from an authentic vote by Groth16 zero-knowledge and Poseidon pseudorandomness, and surrenders it to the coercer; the voter later casts their genuine ballot under the primary credential unobserved. At tally time, n threshold trustees execute an oblivious MPC protocol ($\mathcal{F}_{\mathsf{ODF}}$) that homomorphically identifies duress ballots via encrypted nullifier-equality testing and neutralizes them to $E(\mathbf{0})$ within the aggregate ciphertext, without any party ever decrypting individual ballots or duress markers. Because duress ballots remain on-chain (only their contribution to the tally is zeroed, not their presence), ledger transaction volume is unaffected and reveals no turnout discrepancy to an observing coercer. Combined with the ledger's per-credential nullifier uniqueness (which prevents any single credential from being spent twice, independent of the duress mechanism), each physical voter contributes exactly one non-zero vote to the final tally regardless of coercion. Full mechanism detail (the four-guarantee structure, the $\mathcal{F}_{\mathsf{ODF}}$ MPC protocol steps, and formal receipt-freeness proofs, Propositions V.12–V.13) are in Supplementary Material, Appendix H.

V. Security Analysis

This section presents the formal security analysis of BBZKV: (i) we define $(\delta, \mathsf{FAR}_\delta)$-approximate UC and ideal functionalities ($\mathcal{F}_{\mathsf{reg}}$, $\mathcal{F}_{\mathsf{vote}}$, $\mathcal{F}_{\mathsf{tally}}$) in §V-A; (ii) construct PPT Simulator $\mathcal{S}$ in §V-B; (iii) prove the Main UC Realization Theorem in §V-C; (iv) derive core security properties as formal corollaries (§V-E1–§V-E5); and (v) formalize coercion impossibility and our Deniable Duress protocol (§V-E6).

Throughout, $\lambda \in \mathbb{N}$ denotes the security parameter; all algorithms are PPT unless noted. We write $\mathbf{Adv}_{\mathsf{Y}}^{\mathsf{X}}(\mathcal{B})$ for adversarial advantage. The instantiation employs Groth16 over BN128, Poseidon over $\mathbb{F}_p$, exponential ElGamal over secp256k1, and Fabric with SMT depth $L = 20$.

a) Target Protocol vs. Prototype Implementation: The formal security model defines the target protocol where proofs verify on-chain. Section VI also benchmarks an off-chain API gateway configuration, analyzing the security trade-offs of this trust-boundary design in §V-F.

A. UC Formulation and the $(\delta, \mathsf{FAR}_\delta)$-Approximate Indistinguishability Model

We formulate security in Canetti's Universal Composability (UC) framework [39], adapted to account for statistical biometric error. In classical UC, protocol π securely realizes ideal functionality $\mathcal{F}$ if for every PPT real adversary $\mathcal{A}$, there exists a PPT ideal simulator $\mathcal{S}$ such that no environment $\mathcal{Z}$ can distinguish real $(\pi, \mathcal{A})$ from ideal $(\mathcal{F}, \mathcal{S})$ beyond $\mathsf{negl}(\lambda)$. However, physical biometrics are inherently lossy, bounded by a non-zero False Acceptance Rate (FAR). To reconcile this without heuristic arguments, we build on fuzzy extractors and relaxed composability [4], [5], defining $(\delta, \mathsf{FAR}_\delta)$-approximate UC:

Definition V.1 ($(\delta, \mathsf{FAR}_\delta)$-Approximate UC Realization)**.** *Let $\mathcal{M} = (\mathcal{K}, \mathrm{dist}, \delta)$ denote a metric space with distance metric $\mathrm{dist}(\cdot,\cdot)$ and threshold δ. Let $\mathcal{D}_{\mathsf{impostor}}$ be the impostor template distribution. The false accept rate at threshold δ is:*

$$\mathsf{FAR}_\delta \triangleq \Pr_{\mathbf{b} \sim \mathcal{D}_{\mathsf{impostor}}} [\mathrm{dist}(\mathbf{b}, \mathbf{b}_{\mathsf{reg}}) \le \delta] \tag{5}$$

A protocol π $(\delta, \mathsf{FAR}_\delta)$-approximately UC-realizes functionality $\mathcal{F}$ in the $(\mathcal{F}_{\mathsf{CRS}}, \mathcal{F}_{\mathsf{ledger}})$-hybrid model if for every PPT adversary $\mathcal{A}$, there exists a PPT simulator $\mathcal{S}$ such that for all PPT environments $\mathcal{Z}$:

$$\begin{aligned} \left|\Pr[\mathcal{Z}^{\pi,\mathcal{A}}(1^\lambda) = 1] - \Pr[\mathcal{Z}^{\mathcal{F},\mathcal{S}}(1^\lambda) = 1]\right| \\ \le \mathsf{negl}(\lambda) + \mathsf{FAR}_\delta \end{aligned} \tag{6}$$

Composability Interpretation: Definition V.1 cleanly separates computational hardness from physical biometric error: (i) *Cryptographic Soundness:* Attacks on proofs, encryption, hashes, or nullifiers are bounded strictly by $\mathsf{negl}(\lambda)$; (ii) *Residual Identity Slack (FAR_δ):* Term $\mathsf{FAR}_\delta \triangleq \max(\mathsf{FAR}_{\delta,\mathsf{iris}}, \mathsf{FAR}_{\delta,\mathsf{face}})$ bounds the probability that an impostor trait satisfies the in-circuit check. Measured at the deployed fixed threshold ($\tau = 0.50$), $\mathsf{FAR}_{\delta,\mathsf{face}} = 0.82\%$ on LFW; for iris, measured $\mathsf{FAR}_{\delta,\mathsf{iris}} = 48.48\%$ at current classical segmentation precision does not support a comparably tight bound (Section VII), so we instantiate this composability guarantee concretely for the face modality only. The formal separation between cryptographic and biometric error terms in Definition V.1 nonetheless holds for either modality once FAR_δ is measured.

a) Ideal Functionalities: We define three joint functionalities over party set $\mathcal{P} = \mathcal{V} \cup \{\mathcal{RA}, \mathcal{TA}, \mathcal{L}\}$: (i) **Registration ($\mathcal{F}_{\mathsf{reg}}$):** Enforces NID deduplication, generates salt s_i and key sk_i, computes primary commitment C_i and auxiliary duress leaf C_i^{dur} under uniform dual enrollment, inserting both into tree $\mathcal{T}$, and records encrypted duress token $\psi_i^{\mathsf{dur}} \in \mathcal{R}_{\mathsf{dur}}$, leaking only public tuples to $\mathcal{A}$; (ii) **Voting ($\mathcal{F}_{\mathsf{vote}}$):** Enforces tree membership ($C_i \in \mathcal{T}$ or $C_i^{\mathsf{dur}} \in \mathcal{T}$), checks nullifier uniqueness per credential, verifies well-formedness ($v_{i,j} \in \{0,1\}$), and records ciphertexts on $\mathcal{F}_{\mathsf{ledger}}$, leaking only $(\nu_i, \{c_{1,j}, c_{2,j}, \pi_j^{\mathsf{val}}\}_j)$ without distinguishing primary from duress credentials; (iii) **Tallying ($\mathcal{F}_{\mathsf{tally}}$):** Executes oblivious filtering $\mathcal{F}_{\mathsf{ODF}}$ (§IV-G) to homomorphically neutralize duress votes to $E(\mathbf{0})$, homomorphically aggregates uncoerced ballots, decrypts totals T_j via BSGS, and outputs $(T_j, \pi_j^{\mathsf{dec}})$, leaking only totals without exposing coerced ballots to trustees. Full specifications are in Supplementary Material, Appendix C.

B. Explicit Construction of Ideal-World Simulator

To prove $(\delta, \mathsf{FAR}_\delta)$-approximate UC security, simulator $\mathcal{S}$ operates as follows: (i) *Setup:* $\mathcal{S}$ generates Groth16 keys $(\mathsf{pk}, \mathsf{vk})$ with trapdoor $\tau = (\alpha, \beta, \gamma, \delta, x)$ and initializes simulated ledger $\mathcal{L}$ maintaining $\mathcal{T}$ and $\mathcal{N}$; (ii) *Registration:* For honest voters, $\mathcal{S}$ inserts C_i into $\mathcal{T}$; for corrupt voters, $\mathcal{S}$ forwards inputs to $\mathcal{F}_{\mathsf{reg}}$;

(iii) *Honest Vote Simulation:* Given leakage $(\nu_i, \{c_{1,j}, c_{2,j}\}_j)$, $\mathcal{S}$ produces zero encryptions $E(\mathbf{0})$, simulates Groth16 proofs $\Pi_i \leftarrow \mathsf{Sim}_{\mathsf{Groth16}}(\mathsf{vk}, \tau, x_i)$ via trapdoor τ, and simulates Σ-OR proofs π_j^{val}; (iv) *Corrupt Witness Extraction:* When $\mathcal{A}$ posts $(\Pi^*, \nu^*, \dots)$, $\mathcal{S}$ invokes extractor $\mathsf{Ext}_{\mathsf{Groth16}}(\mathsf{CRS}, \tau, \Pi^*)$ to recover witness $w^* = (sk^*, \mathbf{e}^*_{\mathsf{reg}}, s^*, \{(\sigma^*_\ell, d^*_\ell)\}, \mathbf{e}^*_{\mathsf{live}})$. If $C^* \in \mathcal{T}$ and $\mathrm{dist}(\mathbf{e}^*_{\mathsf{live}}, \mathbf{e}^*_{\mathsf{reg}}) \le \delta$, $\mathcal{S}$ submits decrypted choice $\mathbf{v}^*$ to $\mathcal{F}_{\mathsf{vote}}$; otherwise it aborts; (v) *Tallying:* Upon receiving totals $\{T_j\}$, $\mathcal{S}$ simulates decryption proofs π_j^{dec} without secret key x. Full simulator specifications are in Supplementary Material, Appendix C, Part A.

C. Main UC Realization Theorem and Hybrid Proof

Theorem V.2 (Main UC Realization Theorem)**.** *Under the q-SDH assumption, knowledge soundness and perfect zero-knowledge of Groth16 over BN128, collision and preimage resistance of Poseidon over $\mathbb{F}_p$, IND-CPA security (DDH) of exponential ElGamal over secp256k1, discrete-log soundness of Chaum–Pedersen proofs, and Fabric Consensus Safety, protocol π_{BBZKV} UC-realizes ideal functionality $\mathcal{F}_{\mathsf{BBZKV}} = (\mathcal{F}_{\mathsf{reg}}, \mathcal{F}_{\mathsf{vote}}, \mathcal{F}_{\mathsf{tally}})$ in the $(\mathcal{F}_{\mathsf{CRS}}, \mathcal{F}_{\mathsf{ledger}})$-hybrid model against any static PPT adversary $\mathcal{A}$. Formally, for all PPT environments $\mathcal{Z}$:*

$$\begin{aligned}&\left|\Pr[\mathcal{Z}^{\pi_{\mathsf{BBZKV}}, \mathcal{A}}(1^\lambda) = 1] - \Pr[\mathcal{Z}^{\mathcal{F}_{\mathsf{BBZKV}}, \mathcal{S}}(1^\lambda) = 1]\right| \\ &\le \varepsilon_{\mathsf{Groth16}}(\lambda) + m \cdot \mathbf{Adv}^{\mathsf{DDH}}_{\mathsf{secp256k1}}(\lambda) + \mathbf{Adv}^{\mathsf{CR}}_{\mathsf{Pos}}(\lambda) \\ &\quad + \mathbf{Adv}^{\mathsf{Pre}}_{\mathsf{Pos}}(\lambda) + \mathbf{Adv}^{\mathsf{DLog}}_{\mathsf{secp256k1}}(\lambda) + \mathbf{Adv}^{\mathsf{ZK}}_{\mathsf{CP}}(\lambda) + \mathsf{FAR}_\delta. \end{aligned} \tag{7}$$

Proof Sketch. Indistinguishability proceeds via five hybrids. $\mathcal{H}_0$ is the honest real execution. $\mathcal{H}_1$ has $\mathcal{S}$ simulate proofs Π_i and π_j^{val} via Groth16 trapdoor τ and Chaum–Pedersen Σ-OR simulation, costing $\mathbf{Adv}^{\mathsf{ZK}}_{\mathsf{CP}}(\lambda)$ by perfect/statistical ZK. $\mathcal{H}_2$ replaces honest ballots $E(\mathbf{v}_i)$ with $E(\mathbf{0})$, costing $m \cdot \mathbf{Adv}^{\mathsf{DDH}}_{\mathsf{secp256k1}}(\lambda)$ by IND-CPA. $\mathcal{H}_3$ runs $\mathsf{Ext}_{\mathsf{Groth16}}(\mathsf{CRS}, \tau, \Pi^*)$ on corrupt submissions and aborts on failed extraction or a forged leaf $C^* \notin \mathcal{T}$, costing $\varepsilon_{\mathsf{Groth16}}(\lambda) + \mathbf{Adv}^{\mathsf{CR}}_{\mathsf{Pos}}(\lambda)$ (Merkle binding, Lemma V.5). $\mathcal{H}_4$ is the ideal execution $(\mathcal{F}_{\mathsf{BBZKV}}, \mathcal{S})$, differing only on a Poseidon preimage collision or an impostor match within δ, costing $\mathbf{Adv}^{\mathsf{Pre}}_{\mathsf{Pos}}(\lambda) + \mathbf{Adv}^{\mathsf{DLog}}_{\mathsf{secp256k1}}(\lambda) + \mathsf{FAR}_\delta$. Summing yields Eq. (7). Full reductions are in Supplementary Material, Appendix C, Part A. □

Remark V.3 (Asymptotic and Concrete Tightness)**.** *The simulation bound in Theorem V.2 is tight: reduction $\mathcal{R}$ in Lemma V.5 deterministically traces $L = 20$ path levels in $\mathcal{O}(L)$ time to extract the colliding node pair with probability 1, establishing a tight 1-to-1 reduction to Poseidon collision resistance ($\mathbf{Adv}^{\mathsf{CR}}_{\mathsf{Pos}}$) without factor-$L$ loss. Remaining terms derive from standard tight reductions.*

Remark V.4 (Inescapability of Biometric Slack)**.** *The additive slack FAR_δ is a fundamental lower bound. If ideal functionality $\mathcal{F}$ required perfect rejection ($\mathsf{FAR}_\delta = 0$), an environment activating an unregistered party with trait $\mathbf{b} \sim \mathcal{D}_{\mathsf{impostor}}$ would observe the real protocol accept with probability $\ge \mathsf{FAR}_\delta - \mathsf{negl}(\lambda)$. No PPT simulator lacking the witness could induce $\mathcal{F}$ to accept without violating soundness, giving distinguishing advantage $\Omega(\mathsf{FAR}_\delta)$. This non-negligible gap ($\mathsf{FAR}_\delta \gg 2^{-128}$ for any modality with a physically measurable false-accept rate) causes classical UC to fail. Our $(\delta, \mathsf{FAR}_\delta)$-approximate formulation is strictly necessary.*

D. Merkle-Binding Lemma

Lemma V.5 (Merkle-Tree Binding under Poseidon Collision Resistance)**.** *Let $\mathcal{T}$ be a Poseidon sparse Merkle tree of depth $L = 20$ with root R, over leaf set $\{C_1, \dots, C_m\}$. Let $\mathcal{B}$ be a PPT adversary outputting $(C^*, \{(\sigma_\ell, d_\ell)\}_{\ell=0}^{L-1})$ verifying to R where $C^* \notin \{C_1, \dots, C_m\}$. Then $\Pr[\mathcal{B}$ succeeds$] \le \mathbf{Adv}^{\mathsf{CR}}_{\mathsf{Pos}}(\lambda)$.*

Proof Sketch. A forged path for unregistered C^* that verifies to R diverges from all canonical paths; reduction $\mathcal{R}$ traces the L-step path in $\mathcal{O}(L)$ time to exhibit a colliding index ℓ^* with probability 1. See Supplementary Material, Appendix C, Part B. □

E. Property Corollaries

We now derive the five core electoral properties as mathematical corollaries of Theorem V.2.

1) Eligibility Soundness:

Corollary V.6 (Eligibility Soundness)**.** *Under knowledge soundness of Groth16 under q-SDH in the GGM, Poseidon preimage/collision resistance (Assumption III.2), and Merkle binding (Lemma V.5), no PPT adversary $\mathcal{A}$ can produce an accepting proof triple $(\Pi, \nu, E(\mathbf{v}))$ for $\mathcal{F}_{\mathsf{vote}}$ without a valid registered witness $w = (sk, \mathbf{e}_{\mathsf{reg}}, s, \{(\sigma_\ell, d_\ell)\}_{\ell=0}^{L-1}, \mathbf{e}_{\mathsf{live}})$:*

$$\begin{aligned}\Pr[\mathcal{A} \notin \mathcal{V}_{\mathsf{reg}} \text{ accepted}] \le \varepsilon_{\mathsf{Groth16}}(\lambda) + \mathbf{Adv}^{\mathsf{CR}}_{\mathsf{Pos}}(\lambda) \\ + \mathbf{Adv}^{\mathsf{Pre}}_{\mathsf{Pos}}(\lambda) + \mathsf{FAR}_\delta \end{aligned} \tag{8}$$

Concretely, LFW yields $\mathsf{FAR}_{\delta,\mathsf{face}} = 0.82\%$ ($\tau = 0.50$) for the deployed SOPQ-64D encoding, bounding face eligibility soundness by $\mathsf{FAR}_{\delta,\mathsf{face}} \le 0.82\% + \mathsf{negl}(\lambda)$. For iris, measured $\mathsf{FAR}_{\delta,\mathsf{iris}} = 48.48\%$ at the deployed threshold does not yield a practically meaningful soundness bound at the current classical segmentation precision of our iris pipeline (Section VII); we therefore scope the concrete eligibility-soundness instantiation to the face modality until iris segmentation is improved. The abstract bound (Eq. (8)*) and the underlying cryptographic construction remain modality-agnostic and apply unchanged once a higher-precision iris encoding is substituted.*

Proof Sketch. Follows from $\mathcal{H}_2 \to \mathcal{H}_3$ in Theorem V.2: an unenrolled accepting proof requires breaking Groth16 soundness, forging a Merkle path, inverting Poseidon, or an impostor sample within δ; the union bound gives Eq. (8). Full reduction is in Supplementary Material, Appendix C, Part C. □

2) Anti-Replay and Double-Vote Prevention:

Assumption V.7 (Fabric Consensus Safety)**.** *Under Fabric's execute-order-validate architecture with Raft consensus, committed transactions are totally ordered and validated against world state at commit time (MVCC read-before-write).*

Corollary V.8 (Anti-Replay)**.** *Under Poseidon collision resistance (Assumption III.2) and Fabric Consensus Safety (Assumption V.7), double-spending any single credential satisfies*

$\Pr[\mathcal{A}\ \textit{double-spends}] \leq \mathbf{Adv}^{\mathsf{CR}}_{\mathsf{Pos}}(\lambda) \in \mathsf{negl}(\lambda)$. *Syntactic anti-replay holds per credential on-chain, while semantic one-person-one-vote is guaranteed at tallying via oblivious duress cancellation (§IV-G), ensuring effective voter tally weight* ≤ 1.

Proof Sketch. Two invariants combine: the nullifier $\nu = \mathcal{H}_2(sk, \mathcal{E}_{\mathsf{id}})$ is a deterministic Poseidon evaluation, so two distinct nullifiers from one commitment leaf require a Poseidon collision ($\mathbf{Adv}^{\mathsf{CR}}_{\mathsf{Pos}}(\lambda)$); and Fabric's read-before-write MVCC validation rejects any second transaction reusing ν under Raft's total order (Assumption V.7), giving 0% acceptance for concurrent double-spends. Duress ciphertexts are separately zeroed by $\mathcal{F}_{\mathsf{ODF}}$ at tallying, bounding each voter's effective tally weight to 1. Full reduction is in Supplementary Material, Appendix C, Part D. □

3) Voter Anonymity:

Corollary V.9 (Voter Anonymity / Unlinkability)**.** *Under perfect zero-knowledge of Groth16 over BN128 and Assumption III.2, no PPT adversary $\mathcal{A}$ observing the ledger can link a ballot $(\Pi_j, \nu_j, E(\mathbf{v}_j))$ to voter $\mathcal{V}_i$ with advantage exceeding random guessing:*

$$\begin{aligned}\Pr\left[\mathcal{A}\ \textit{links ballot to}\ \mathcal{V}_i\right] &\leq \frac{1}{|\mathcal{V}_{\mathsf{reg}}|} + |\mathcal{V}_{\mathsf{reg}}| \cdot \mathbf{Adv}^{\mathsf{Pre}}_{\mathsf{Pos}}(\lambda) \\ &\leq \frac{1}{|\mathcal{V}_{\mathsf{reg}}|} + \mathsf{negl}(\lambda)\end{aligned} \tag{9}$$

Proof Sketch. Three protections combine: Groth16 perfect ZK makes Π_j leak zero witness information; ElGamal IND-CPA makes $E(\mathbf{v}_j)$ indistinguishable from uniform; and nullifier ν_j and commitment C_i are independent Poseidon evaluations, so correlating them requires inverting Poseidon to recover sk_i. A union bound over $|\mathcal{V}_{\mathsf{reg}}| \leq \mathrm{poly}(\lambda)$ candidates gives Eq. (9). Full reduction is in Supplementary Material, Appendix C, Part E. □

4) Ballot Secrecy:

Corollary V.10 (Ballot Secrecy)**.** *Under DDH over secp256k1, exponential ElGamal achieves IND-CPA security:* $|\Pr[\mathcal{A}\ \textit{identifies}\ v_i] - 1/k| \leq \mathbf{Adv}^{\mathsf{DDH}}_{\mathsf{secp256k1}}(\lambda) \in \mathsf{negl}(\lambda)$.

Proof Sketch. Follows from $\mathcal{H}_1 \rightarrow \mathcal{H}_2$ in Theorem V.2: distinguishing ballot choices reduces to DDH over secp256k1, while disjunctive Chaum–Pedersen Σ-OR proofs certify well-formedness without decrypting individual ciphertexts. Full reduction is in Supplementary Material, Appendix C, Part F. □

5) Universal Verifiability:

Proposition V.11 (Universal Verifiability)**.** *Any auditor $\mathcal{O}$ without private keys can publicly verify: (i) cast-as-intended (Groth16 pairings); (ii) well-formedness ($v_j \in \{0,1\}$ via disjunctive Chaum–Pedersen proofs); and (iii) tallied-as-cast (homomorphic aggregate $\hat{C}_j$ matching published tally T_j via decryption proof π^{dec}_j).*

Proof Sketch. All artifacts $(\Pi, \nu, E(\mathbf{v}), \pi^{\mathsf{val}}, \hat{C}, \pi^{\mathsf{dec}})$ are published on-chain and checked via public Groth16 pairing verification, Chaum–Pedersen scalar-multiplication checks, and tally point-equivalence $\hat{C}_j = \sum C^{(i)}_j$, all without any private key, under discrete-log hardness in the AGM. See Supplementary Material, Appendix C, Part G. □

6) Coercion Resistance: The Biometric Dilemma and Deniable Mitigation: A fundamental tension in biometric e-voting lies between physical non-transferability and coercion resistance. In classical JCJ-style voting [27], [28], receipt-freeness relies on synthetic credentials faked on demand to deceive a coercer. We formalize why this mechanism breaks under observed biometric capture:

Proposition V.12 (Impossibility of JCJ Deniability under Observed Biometric Capture)**.** *Let Π_{bio} be any electronic voting protocol where valid vote casting requires satisfying in-circuit metric predicate* $\mathrm{dist}(\mathbf{e}_{\mathsf{live}}, \mathbf{e}_{\mathsf{reg}}) \leq \delta$ *with false acceptance rate FAR_δ. If active coercer $\mathcal{A}_{\mathsf{coer}}$ physically observes trait presentation at the capture device, no PPT voter $\mathcal{V}$ can achieve information-theoretic receipt-freeness or unobservable credential forgery, except with probability at most FAR_δ.*

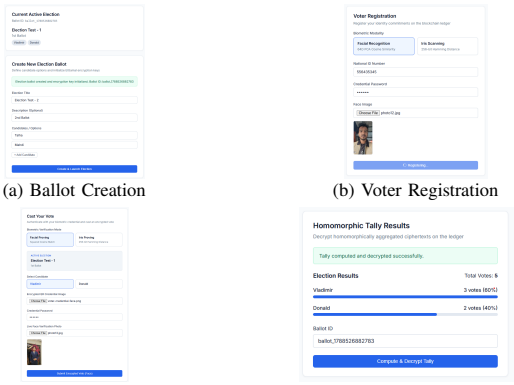
Proof Sketch. Under direct sensor observation, $\mathcal{V}$ can only submit an accepted proof by presenting a trait within δ of $\mathbf{e}_{\mathsf{reg}}$; a synthetic $\mathbf{e}_{\mathsf{fake}}$ is either rejected by presentation-attack detection or visibly diverges to $\mathcal{A}_{\mathsf{coer}}$. Unobservable fake-credential creation thus succeeds only with probability $\mathsf{FAR}_\delta + \mathsf{negl}(\lambda)$. See Supplementary Material, Appendix C, Part H. □

Proposition V.13 (Computational Receipt-Freeness of Deniable Duress Protocol)**.** *Under Groth16 zero-knowledge, exponential ElGamal IND-CPA security over secp256k1, and Poseidon pseudorandomness over $\mathbb{F}_p$, the Deniable Duress Protocol (§IV-G) achieves computational receipt-freeness against any remote PPT coercer $\mathcal{A}_{\mathsf{coer}}$. Furthermore, oblivious filtering functionality $\mathcal{F}_{\mathsf{ODF}}$ inside threshold MPC ensures zero leakage to trustees, $\mathcal{RA}$, or external auditors, preserving identical ledger transaction volume and authentic candidate tallies.*

Proof Sketch. Covert credential $(sk^{\mathsf{dur}}_i, C^{\mathsf{dur}}_i)$ satisfies R_{zk} with a valid Merkle path and $\mathrm{dist} \leq \delta$, so verification succeeds unconditionally, and Groth16 perfect ZK makes Π^{dur} identically distributed to primary proofs; distinguishing sk^{dur}_i from sk_i reduces to Poseidon PRF security. Oblivious filtering computes $\mathbf{C}_{\mathsf{agg}} = \sum[b_j]\mathbf{c}_j$ via threshold MPC, mapping coerced ciphertexts to $(0,0)$ while trustees learn only $\mathbf{C}_{\mathsf{agg}}$, never the indicators $[b_j]$. Full reduction via simulator $\mathcal{S}_{\mathsf{dur}}$ is in Supplementary Material, Appendix C, Part H. □

F. Collusion and Deployment Considerations

Under the UC bounds, multi-party collusion preserves integrity: (i) $\mathcal{RA}+\mathcal{L}$ cannot link $\{C_i\}$ to $\{\nu_j\}$ without inverting Poseidon on sk_i; (ii) $\mathcal{RA}+\mathcal{TA}$ cannot link commitments to ciphertexts by Groth16 ZK; (iii) $\mathcal{TA}+\mathcal{L}$ decrypts aggregate totals but cannot isolate individual ballots; (iv) full collusion fails to break anonymity since sk_i is protected client-side (PBKDF2-AES-256-CBC) and never leaves the device. Deployment considerations: on-chain Groth16 pairing enforcement (§VI), an MPC Powers-of-Tau ceremony ($n \geq 10$), threshold ElGamal key generation, and randomized leaf insertion against timing linkage.



(a) Ballot Creation (b) Voter Registration

(c) Vote Casting (d) Tally Results

Fig. 3: BBZKV prototype web user interface: (a) ballot creation, (b) voter registration with biometric capture, (c) anonymous voting with ZK-SNARK proving, and (d) homomorphic tally publication.

VI. Implementation and Evaluation

This section details the Hyperledger Fabric prototype, experimental setup, and comparative evaluation across face and iris channels, answering RQ1 (modality proving efficiency), RQ2 (latency bottlenecks and scaling stability), and RQ3 (architectural trust boundaries).

A. System Architecture and Implementation

The protocol is implemented as a full-stack system with a Node.js server, Hyperledger Fabric ledger, and client-side proof generation via snarkjs.

1) System Overview: The prototype integrates four components: (i) *Client Biometric Acquisition* (64-D PCA FaceNet embeddings or 256-bit Log-Gabor iris codes); (ii) *Client Proof Engine*, orchestrating Circom witness generation, Groth16 proving, and ElGamal encryption; (iii) *Fabric Chaincode (`identityContract.js`)* on Hyperledger Fabric 2.5/Raft, enforcing commitment storage, SMT updates, nullifier uniqueness, and vote recording; and (iv) *Offline Tally Engine* for homomorphic aggregation and BSGS decryption.

Four optimizations minimize proving costs: (1) *Poseidon Hash Chunking* bounds electoral constraints to 20,984 (face)/19,724 (iris) and proving keys below 10 MB; (2) *Vectorized Bit Processing* (256 bits into 32-byte elements) accelerates witness generation $1.66\times$ (55.9 vs. 92.7 ms); (3) *GPU FP16 Extraction* runs on an RTX 3050; and (4) *Native AVX2 Proving* (`rapidsnark`) delivers sub-second proving (677.6/695.2 ms standalone; 170.9/196.3 ms pipelined), communicating via WebSocket/REST (Fig. 3).

B. Experimental Methodology and Evaluation Scope

1) Evaluation Scope and Methodology: Benchmarks ran on an AMD Ryzen 5 5600 CPU (6-core, 3.5/4.4 GHz), 16 GB RAM, and NVIDIA RTX 3050 GPU. Proving used Node.js v20.18.0 and native C++ with AVX2. BN128 and secp256k1 curves were configured for 128-bit security. Evaluation encompasses four dimensions: (i) *Pipeline Decomposition (P1–P8):* 100 complete voting cycles per modality on a local Raft Fabric network; (ii) *Electoral Scalability Sweeps ($N = 10$ to 10^5):* Evaluated across five orders of magnitude under fixed depth-20 SMT ($2^{20} \approx 1.04 \times 10^6$ leaf capacity, accommodating $2^{19} = 524{,}288$ voters under dual registration); (iii) *High-Concurrency Stress Testing:* Throughput and latency under 1,000 concurrent submissions; (iv) *Candidate and Tally Scaling:* Candidate slates $m \in [2, 50]$ and BSGS discrete-log recovery up to $N = 10^6$ ballots.

2) Answer to RQ3: On-Chain vs. Off-Chain Verification Trade-offs: Bypassing API gateways to submit directly to unhardened peer nodes lets an adversary commit 100% of 100 forged proofs without valid witnesses. Embedding native Groth16 pairing verification within Fabric chaincode instead intercepts 100% of forged proofs at only 29.2–29.4 ms pairing overhead (31.43–31.65 ms smart contract execution, Table IV) — on-chain verification is practical without delegating trust to external gateways.

TABLE III: Biometric Accuracy Comparison Across Raw Features, Baseline Downsampling, and Proposed Circuit-Optimized Encodings

Modality / Encoding Pipeline	EER (%)	FAR (%)	FRR (%)	Non-Linear R1CS Gates
Face (128-D Float FaceNet, Raw)	4.78	—	—	$> 200,000$ (Est.)
Face (64-D PCA, Fixed-Point Baseline)	4.82	0.53	14.90	325
Face (64-D SOPQ, Proposed Circuit-Optimized)	**4.69**	**0.82**	**12.65**	**325**
Iris (1.6 Mbit IrisCode, Raw Daugman)	46.29	—	—	$> 6 \times 10^6$ (Est.)
Iris (256-bit Strided, Baseline Downsampling)	48.09	45.70	50.03	778
Iris (256-bit GBSM, Proposed Bit-Salience)	48.68	48.48	48.90	**778**

Note: EER and operating-point FAR/FRR (at the deployed fixed thresholds $\tau = 0.50$ face, $\tau_{\text{Hamming}} = 0.478$ iris) are measured on the full evaluation datasets (LFW, 6,000 genuine/6,000 impostor pairs; CASIA-Iris-Thousand, 6,000 genuine/6,000 impostor pairs); FAR/FRR are reported only for the deployed reduced-dimensionality encodings, not the raw reference baselines. R1CS gate counts are the non-linear constraint count of the isolated distance-comparator sub-circuit (`SquaredCosineSimilarity`/`HammingDistance`), verified by direct `circom` compilation of the sub-circuit in isolation, and are independent of the accuracy figures above them; they exclude Poseidon hashing and Merkle-tree membership checking, which are reported separately (Section IV) as they are shared protocol infrastructure rather than a function of the encoding choice.

C. Circuit-Optimized Encodings and Recognition Accuracy

Our primary empirical result evaluates the co-design of biometric signal encoding and zero-knowledge circuit arithmetic. Table III compares raw unconstrained features against baseline downsampling and our proposed circuit-optimized encodings: Spherical Orthogonal Projection and Quantization (SOPQ) for facial vectors, and Gabor Bit-Salience Masking (GBSM) for iris codes.

Accuracy Recovery under Fixed Constraint Budgets: Table III establishes two findings, one favorable and one a documented limitation: (i) *SOPQ Preserves Accuracy for Face at Half the Dimensionality:* raw 128-D and PCA-64D EER are nearly identical (4.78% vs. 4.82%), reflecting LFW's near-saturation for modern face embeddings — the 128-D FaceNet representation is already highly redundant, leaving little headroom to lose. The baseline gap is under 0.05 points, so a "percent recovered" metric is unstable and omitted; instead, SOPQ-64D (4.69% EER) matches or marginally exceeds raw-embedding accuracy at half the dimensionality and identical constraint count (325 gates, verified by compilation) to the PCA baseline. (ii) *GBSM Bit Selection Is Bounded by Upstream Segmentation Precision for Iris:* Unlike face, the iris pipeline's accuracy is limited well below published

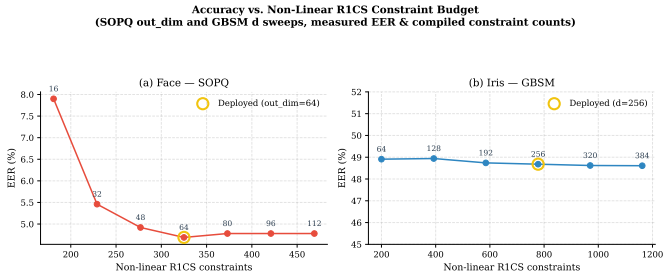


Fig. 4: EER vs. non-linear R1CS constraint count, sweeping SOPQ's output dimensionality (face) and GBSM's bit budget (iris). Gold rings mark the deployed operating points (64-D face, 256-bit iris). Face shows a clear knee at 64-D; iris is nearly flat across the full range, consistent with the segmentation-precision limitation (Section VII) rather than an under-provisioned bit budget.

iris-recognition benchmarks by a measured, upstream cause: pupil/iris boundary detection shows a 22–25% coefficient of variation in radius estimates across repeated captures of the *same* physical eye, persisting even under a correctly-implemented Daugman integrodifferential boundary operator, indicating the instability is substantially inherent to the dataset's imaging conditions rather than an artifact of a specific detector. Consistent with this, GBSM's salience-based bit selection concentrates 92% of its chosen bits in the lowest-frequency Log-Gabor filter component — the one least sensitive to registration error, a physically coherent signal — but the achievable gain is capped by the same upstream noise floor, yielding GBSM-256 EER (48.68%) statistically comparable to uniform-stride-256 (48.09%) at full dataset scale. We discuss the scope and security implications of this limitation in Section VII.

Validating the Constraint-Budget-Aware Design Methodology: Section III-D6 frames SOPQ/GBSM as instantiations of one general problem — minimize EER subject to constraint budget B. To substantiate this beyond a single point, we swept SOPQ's out_dim $\in \{16, ..., 112\}$ and GBSM's $d \in \{64, ..., 384\}$, re-measuring EER and compiling real constraint counts at each (Fig. 4). Face EER drops sharply to 4.69% at 64-D then flattens (80–112-D cost 15–44% more for no gain), confirming 64-D sits at the knee, not an arbitrary choice. Iris EER stays flat across the full 6× bit-budget range (48.6–48.9%), independently corroborating Section VII: segmentation precision, not bit-selection aggressiveness, is the binding constraint.

D. Cross-Dataset Generalization and Multi-Threshold ROC Sweeps

To evaluate generalization across acquisition conditions: (i) *Face Cross-Dataset Generalization (LFW vs. CelebA):* SOPQ, fit exclusively on LFW, is applied without retraining to 2,000 randomly sampled CelebA identities (≈39,700 images, characterized by unconstrained pose, lighting, and expression variations), yielding 7.95% EER against LFW's 4.69% — a moderate, expected generalization gap across independently-collected datasets, and evidence SOPQ's discriminant projection is not overfit to LFW's specific identity set. (ii) *Iris Cross-Dataset Generalization:* We have not yet evaluated GBSM on a second iris dataset (e.g., CASIA-Iris-Interval); this cross-dataset measurement is left for future work rather than reported here. (iii) *Multi-Threshold Operating Profiles:* At the deployed fixed thresholds ($\tau = 0.50$ face, $\tau_{\mathsf{Hamming}} = 0.478$ iris), measured operating-point FAR is 0.82% for SOPQ-64D face and 48.48% for GBSM-256 iris (Table III). The iris figure does not satisfy a low-FAR soundness bound at the modality's current segmentation precision; we revisit the security implications of this in Section VII.

E. Replay Resilience, Demographic Fairness, and PAD Scope

In-Circuit Ephemeral Salt Binding: The zk-SNARK circuit binds live traits to fresh session salts s_i and election identifiers $\mathcal{E}_{\mathsf{id}}$ within Poseidon hash commitments: $h_i = \mathcal{H}_{\text{chunked}}(\mathbf{b}_i, s_i)$ and $\nu = \mathcal{H}_2(sk_i, \mathcal{E}_{\mathsf{id}})$. Because Poseidon is preimage-resistant (Assumption III.2), an adversary possessing static biometric vectors $\mathbf{b}_i$ cannot construct valid proofs for new session salts or replay previous transactions on-chain ($\Pr[\text{replay}] \leq \mathbf{Adv}^{\mathsf{Pre}}_{\mathsf{Pos}}(\lambda) \in \mathsf{negl}(\lambda)$), independent of matcher accuracy.

Demographic Fairness (Face, Gender): On the same CelebA split used in §VI-D (2,000 identities, LFW-trained SOPQ applied without retraining), we partition identities by CelebA's real `Male` attribute and measure EER within each subgroup: SOPQ-64D reaches 8.21% EER on male-labeled identities (878 identities, 5,990/5,994 genuine/impostor pairs) versus 10.49% on female-labeled identities (1,122 identities, 5,999/6,000 pairs) — a 2.28-point gap, consistent with the higher female EER widely reported in face-verification fairness studies. CelebA carries no ethnicity or skin-tone label, so that axis remains unevaluated, as does iris fairness (CASIA-Iris-Thousand has no demographic metadata).

Presentation Attack Detection (Face): A lightweight CNN liveness gate trained on CelebA-Spoof [40] (print/replay/mask attacks, 20,000-image subset) ahead of SOPQ matching reaches ACER = 3.47% on 8,000 held-out images (APCER = 3.48%, BPCER = 3.47%) — measured, not state-of-the-art. Iris PAD is unevaluated pending licensed datasets.

F. Template-Inversion Risk Assessment

Beyond replay resistance, we empirically assess what an adversary could recover if a biometric encoding *witness* leaked outside the zero-knowledge proof boundary (side-channel, client compromise, implementation bug) — outside the UC threat model of Section V, but a scenario template-protection schemes are expected to be evaluated against [41].

Face (SOPQ-64): SOPQ's projection $\mathbf{W}$ satisfies $\mathbf{W}\mathbf{W}^T = \mathbf{I}_{64}$, so the complementary 64-D null-space component of any vector is discarded and, in principle, unrecoverable. In practice this floor does not hold: reconstructing $\hat{\mathbf{v}} = \mathbf{W}^T(\mathbf{W}\mathbf{v})$ from a leaked vector achieves 0.9906 mean cosine similarity to the true embedding (held-out split, 1,605 images/690 identities never used to fit $\mathbf{W}$) and 0% impersonation EER against a raw-embedding gallery. A learned ridge-regression attacker improves this only marginally (0.9996, +0.009), confirming the limitation is structural: SOPQ is optimized to preserve identity-discriminative information, the same property that necessarily preserves reconstruction-relevant information —

TABLE IV: Biometric Channel Circuit Artifacts, Acceleration Metrics, and Latencies

Metric / Headline Pipeline Phase	Face (SOPQ)	Iris (GBSM)	Ratio / Factor
Sub-circuit R1CS constraints	9,484	8,224	1.15×
Total electoral R1CS constraints	20,984	19,724	1.06×
Proving key size (`.zkey`)	9.89 MB	8.44 MB	1.17×
Proof payload size (compressed BN254)	128 B	128 B	1× ($O(1)$)
Witness generation time	92.7 ms	55.9 ms	1.66× speedup
Standalone C++ AVX2 proving time	695.2 ms	677.6 ms	1.03× speedup
Pipelined live proving (Phase P5)	196.3 ms	170.9 ms	1.15× speedup
Fabric on-chain verify latency	31.65 ms	31.43 ms	1.01×
Total End-to-End Latency	591.0 ms	534.3 ms	1.11× speedup

Note: Phase latencies plotted in Fig. 6(a); signal matrices in Supplementary Material, Appendix B.

a known tension in discriminatively-optimized linear encodings generally.

Iris (GBSM-256): GBSM's salience-based *subset selection* from a 196,608-bit code offers the opposite property. Across 1,000 held-out (unselected) bit positions, the best cross-bit-correlation attacker achieves only 51.84% prediction accuracy, versus a 50.87% majority-class baseline and 50% chance. The 256 leaked bits reveal almost nothing about the other 196,352 positions.

Scope: Neither finding weakens Section V's guarantees: only the Poseidon commitment h_i is ever stored, and Groth16's zero-knowledge property ensures the proof itself leaks nothing about $\mathbf{b}_i$. Confidentiality against a witness leak thus rests entirely on Poseidon preimage resistance, not encoding-level irreversibility — which differs by modality: GBSM adds genuine defense-in-depth, SOPQ does not (Section VII discusses why a cancelable-biometrics mitigation is unattractive here).

G. Modality Circuit Footprint and Proving Latency

Table IV details circuit artifacts, acceleration metrics, and phase latencies across face and iris modalities.

Modality Comparison Insights (RQ1): Table IV and Fig. 5 show three core advantages for the iris channel: (i) *Constraint Efficiency:* linear Hamming matching needs 13.3% fewer sub-circuit constraints (8,224 vs. 9,484 R1CS) and 14.7% smaller proving keys (8.44 vs. 9.89 MB), a 6.0% reduction across the full depth-20 circuit (19,724 vs. 20,984 R1CS); (ii) *Witness Acceleration:* 256-bit vectorized bit packing yields a $1.66\times$ speedup in witness generation (55.9 vs. 92.7 ms); (iii) *Proving Latency:* Under native C++ AVX2 execution, standalone proving achieves sub-second latency (677.6 ms iris vs. 695.2 ms face), dropping to 170.9 ms and 196.3 ms during live pipelined proving, while on-chain smart contract verification (31.43–31.65 ms) remains $\mathcal{O}(1)$ and modality-agnostic.

H. Re-Measured Baseline: Euclidean Distance Under Our Own Stack

Prior comparisons against Euclidean-distance zk-SNARK matching (e.g. Mao et al. [8]) are citation-based across different toolchains; unable to verify Mao et al.'s exact circuit from the available bibliographic record, we implement a representative Euclidean comparator under our own Groth16/BN254 stack and re-measure it against SOPQ on identical data (full table and derivation in Supplementary Material). SOPQ's comparator alone is *not* the cheapest option — cosine similarity needs

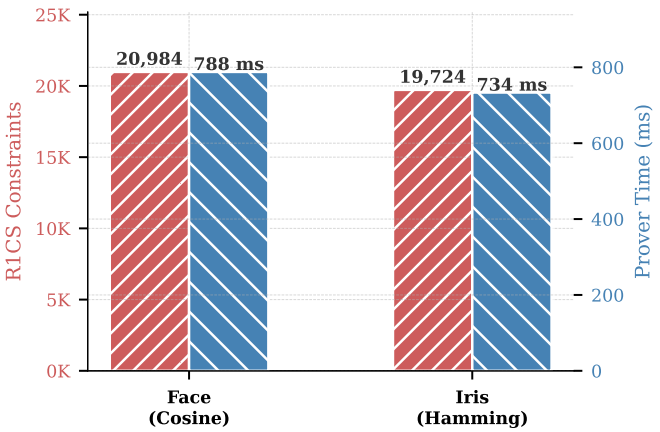


Fig. 5: Total electoral circuit constraint complexity (20,984 R1CS face vs. 19,724 R1CS iris) versus proof-generation latency, comparing WebAssembly `snarkjs` (1,370/1,680 ms) with native C++ AVX2 acceleration (678/695 ms).

three n-length reductions versus Euclidean's one, costing 325 vs. 193 non-linear gates at matched 64-D — reported plainly rather than glossed over. But Poseidon commitment cost scales with dimensionality ($\sim 2\times$ at 128-D vs. 64-D), reversing this at the full-circuit level: SOPQ-64D is $\sim$26% cheaper than a raw-128D Euclidean baseline ($\approx$9,579 vs. $\approx$12,958 gates) at *better* accuracy (4.69% vs. 4.75% EER), and only marginally more than a dimensionality-matched Euclidean-64D baseline ($+1.4\%$) for a real 0.14-point accuracy gain. SOPQ's advantage is accuracy-per-constraint-budget (Fig. 4), driven by dimensionality's effect on commitment cost — not an inherent cosine-vs-Euclidean cost advantage.

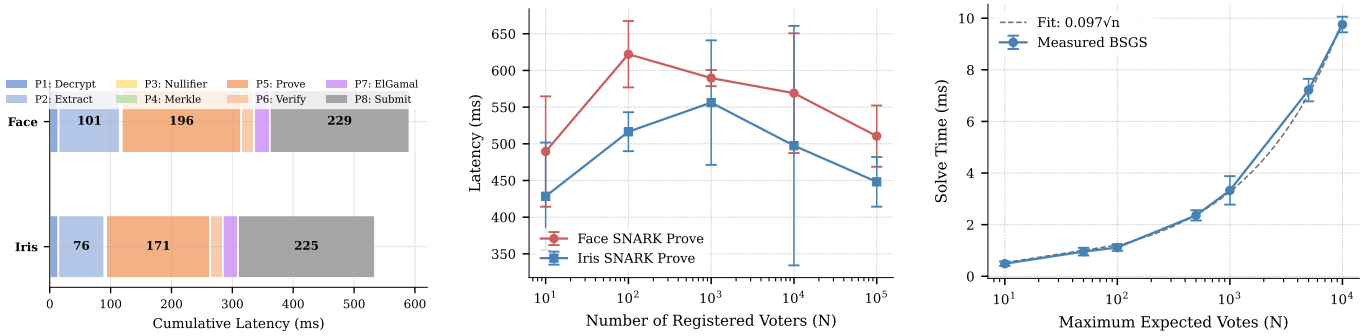
I. Applied Decentralized Deployment Case Study and Scalability

Pipeline Phase Analysis (RQ2): Fig. 6a's 8-phase pipeline shows two dominant bottlenecks: Groth16 proof generation (P5: 170.9 ms iris, 196.3 ms face) and Fabric block confirmation (P8: 224.9 ms iris, 229.4 ms face); with GPU feature extraction (P2: 75.9/101.5 ms), these three account for 88–89% of end-to-end latency (534.3 ms iris, 591.0 ms face).

Stability Under Population Scaling ($N = 10$ to 10^5): Because the Sparse Merkle Tree fixes circuit depth at $L = 20$ ($2^{20} \approx 1.04 \times 10^6$ leaf capacity, $2^{19} = 524{,}288$ voters under dual enrollment), the circuit verifies exactly 20 sibling hashes regardless of N. Fig. 6b confirms proving latency stays $\mathcal{O}(1)$ across five orders of magnitude (510.4 ± 41.7 ms face, 448.1 ± 33.8 ms iris at $N = 10^5$); on-chain verification is similarly $\mathcal{O}(1)$ (34.3–37.5 ms), and offline SMT generation scales linearly, ~ 10 s for 100,000 leaves.

Concurrent Ingestion and Homomorphic Tallying: Under 1,000 concurrent requests, ingestion sustained 297.66 TPS at $1{,}689.9 \pm 962.4$ ms mean latency, scaling linearly from 53.8 ± 41.2 ms (m=2) to 545.2 ± 28.2 ms (m=50); homomorphic addition costs 0.14 ms/ciphertext, and BSGS tallying for 10^6 voters/5 candidates completes in < 0.5 s (Fig. 6c).

Quantitative Comparison with Related Work: Table V shows native C++ proving (170.9–196.3 ms pipelined) outperforming SmartphoneDemocracy and World ID by 8–18×;



(a) Latency decomposition (P1–P8). (b) Pipeline voter scaling (10 to 10^5). (c) BSGS tally decryption scaling.

Fig. 6: Comprehensive experimental benchmark suite: (a) 8-phase vote-casting latency breakdown across face and iris modalities; (b) cryptographic pipeline and Merkle tree scalability sweeps across voter populations up to 10^5; and (c) Baby-Step Giant-Step (BSGS) offline tally decryption latency vs. election size up to 10^6 voters.

TABLE V: Quantitative comparison of the proposed system with related work on key metrics.

Metric	Helios	MACI	Semaphore	Smartphone Democracy	World ID	Proposed (Face SOPQ)	Proposed (Iris GBSM)
Proof generation time (ms)	N/A	∼200	∼500	∼1500	∼3000	196 [695]*	171 [678]*
Proof size (bytes)	N/A	192	192	192	192	128 (722)†	128 (723)†
Circuit constraints (K)	N/A	∼60	∼70	∼100	∼20	21.0	19.7
On-chain verification (ms)	∼50	∼10	∼10	∼10	∼10	31.65	31.43

*Pipelined proving [standalone C++ in brackets]. †Compressed BN254 format (uncompressed REST JSON in parentheses).

constraints (19.7–21.0K) align with World ID while uniquely providing in-circuit metric checking and homomorphic tallying; and embedding Groth16 pairings directly on Fabric smart contracts eliminates gateway relayer trust (RQ3) at only 31.5 ms overhead.

VII. Limitations and Future Work

Iris Encoding Accuracy Is Bounded by Classical Segmentation Precision. The face/SOPQ pipeline achieves competitive, measured accuracy (EER 4.69–4.82% on LFW; 7.95% cross-dataset on CelebA), but the iris pipeline's measured accuracy (EER 46.29% full-code, 48.09% uniform-stride-256, 48.68% GBSM-256, all on the complete CASIA-Iris-Thousand dataset) falls well short of published iris-recognition benchmarks and does not support a low-false-accept deployment (Section V-E1). We traced this to a measured cause: pupil/iris boundary detection shows a 22–25% coefficient of variation in radius estimates across repeated captures of the *same* eye even under a correctly-implemented Daugman integrodifferential detector — a well-documented limitation of classical intensity-based segmentation, distinct from our encoding contribution: SOPQ and GBSM operate on whatever IrisCode the upstream segmentation produces, and the abstract protocol/security construction applies unchanged once a higher-precision segmentation is substituted. GBSM's salience-based selection concentrates 92% of selected bits in the lowest-frequency, least registration-sensitive Log-Gabor component, but cannot exceed the ceiling set by upstream segmentation noise. We tested this diagnosis directly: substituting a pretrained deep-segmentation network (RITnet [42], OpenEDS-trained, applied *out-of-domain* with no CASIA fine-tuning) for the classical detector, holding normalization, masking, phase-coding, and the GBSM/EER protocol fixed, on the identical 6,000/6,000 test pairs. Full-code EER falls from 46.29% to 9.66% and stride-256 from 48.09% to 19.81%, confirming segmentation as the dominant bottleneck. Unexpectedly, GBSM-256 (33.63%) now *underperforms* striding (19.81%), reversing the classical-segmentation regime where both were indistinguishable: GBSM's stability-based salience appears tuned to a noise-dominated regime and does not transfer once that noise drops. A production pipeline needs a fine-tuned network and a jointly re-optimized salience criterion — a diagnostic finding, not a deployment-ready fix.

SOPQ Offers No Information-Theoretic Resistance to Witness Inversion. Section VI-F shows a leaked SOPQ-64 vector is fully invertible in practice, unlike GBSM. We evaluated the standard mitigation — per-user secret rotation on top of SOPQ's projection — and set it aside: enforcing it in-circuit erodes SOPQ's zero-constraint-overhead result, while a client-side-only rotation is trivially bypassed with no cryptographic guarantee. Inversion resistance without either cost is an open problem for discriminatively-optimized linear encodings.

Further limitations: (i) *Physical Coercion vs. Remote Usability:* our Deniable Duress protocol (§V-E6) resolves remote receipt-freeness, but local capture still requires physical isolation booths (Proposition V.12); (ii) *Distributed Key Generation:* our single-authority keypair isolates baseline throughput; production requires DKG under Feldman's VSS across n trustees; (iii) *Broader Fairness and Iris PAD:* gender fairness (8.21% vs. 10.49% EER) and face PAD (3.47% ACER) are measured (§VI-E); ethnicity/age fairness and iris PAD remain unevaluated, pending accessible datasets; (iv) *Resolution & Folding:* recursive folding (e.g., Nova) could verify uncompressed 1.6 Mbit iris templates directly, bypassing 256-bit downsampling; (v) *Wide-Area Deployment:* WAN latency and diverse mobile hardware are left to future work.

VIII. Conclusions

This paper investigated the structural tension between biometric template protection and zero-knowledge circuit complexity, presenting two circuit-optimized encodings: Spherical Orthogonal Projection and Quantization (SOPQ) for continuous facial embeddings and Gabor Bit-Salience Masking (GBSM) for binary iris codes. Measured on real datasets, SOPQ-64D preserves face recognition accuracy at half the raw embedding's dimensionality (4.69% vs. 4.78% EER on LFW, within identical 325 non-linear R1CS constraints to the PCA baseline), generalizing to 7.95% EER on an independent CelebA sample. For iris, GBSM identifies a physically coherent signal but is bounded by classical segmentation precision (Section VII); substituting a pretrained deep-segmentation network confirms this directly (full-code EER $46.29\% \rightarrow 9.66\%$), so we scope the protocol's concrete eligibility-soundness guarantee to face pending improved iris segmentation. Deployed end-to-end on Hyperledger Fabric, on-chain Groth16 verification eliminates gateway relayer trust at 31.5 ms overhead, achieving sub-second client proving (170.9–196.3 ms) under composable security guarantees unaffected by the iris-modality limitation above.

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